

ELEC9731, 2023, Term 1, ASSIGNMENT 1

due – week 7 Tuesday 28/03/2023 17:00(Sydney time).

submission – please submit as **pdf** file by email

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use subject: ELEC 9731-Assignment 1

This assignment should be done analytically, without using computers.

Late assignments will be penalised at 10% of the maximum value per day late

Question 1 (3 marks). Find the rank of the following matrices:

$$A = \begin{pmatrix} -3 & 6 & 3 \\ -2 & 2 & 1 \\ 8 & -4 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} -3 & 0 & 8 \\ -7 & 0 & 14 \\ 3 & 7 & -1 \\ 6 & 4 & -4 \\ 9 & 0 & -3 \end{pmatrix},$$
$$C = \begin{pmatrix} 2 & -3 & 1 & -7 \\ 2 & 3 & 4 & -1 \\ -3 & 4 & 0 & -5 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & -3 \\ 1 & 0 & -1 & -1 \end{pmatrix}.$$

Do they have full rank?

Question 2 (3 marks). For the matrix A given by

$$A = \begin{pmatrix} 7 & 0 & 0 \\ 4 & 3 & 1 \\ 0 & -1 & 3 \end{pmatrix},$$

determine the characteristic equation and the eigenvalues of A .

Question 3 (6 marks). Let $A \in \mathbf{R}^{n \times n}$, $A = (a_{ij})$. Assume that $a_{ij} \geq 0$ for all $1 \leq i, j \leq n$ and $\sum_{i=1}^n a_{ij} = 1$ for all $1 \leq j \leq n$. Prove the following statements:

(i) The matrix A has eigenvalue 1.

(ii) Any eigenvalue λ of A satisfies $|\lambda| \leq 1$.

Question 4 (6 marks). A linear control system is described by the equations

$$\dot{x}(t) = \begin{pmatrix} 1.2 & -4 & 3a \\ 0 & -2 & 0 \\ -4.0 & 3 & 1.6 \end{pmatrix} x(t) + \begin{pmatrix} 0.5 & a-4 & 3.3 \\ 0 & a+5 & 0 \\ 0.2 & a^2 & 3-a \end{pmatrix} u(t),$$

where a varies from $-\infty$ to ∞ . For what values of a this system is

- (i) controllable;
- (ii) stabilizable.

Question 5 (7 marks).

A linear control system is described by the equations

$$\begin{aligned} \dot{x}(t) &= \begin{pmatrix} 1.2 & -2 & 0 \\ 1 & a & 0 \\ 0 & 1 & 0 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} u(t), \\ y(t) &= \begin{pmatrix} 0 & 0 & -1 \end{pmatrix} x(t), \end{aligned}$$

where a varies from $-\infty$ to ∞ .

- (i) Find the characteristic polynomial.
- (ii) Find the range of a for stability of the characteristic polynomial.
- (iii) For $a = 1.5$, find the open loop poles of the system and comment on their stability.
- (iv) For $a = 1.5$, check the controllability of the system.
- (v) For $a = 1.5$, find the modal state space model and re-check controllability using the modal control parameters.
- (vi) For $a = 1.5$, find the transfer function from u to y .